

Instructions.

1. **Due April 10, 11:59 PM uploaded to Blackboard as a single pdf and a single ZIP archive. This deadline is firm. Late uploads will not be accepted.**
2. This project will be done in a team of two members. You may **NOT** discuss or consult with any students outside your team.
3. All work must be typeset in L^AT_EX. Handwritten or scanned submissions will not be accepted.
4. All figures must have axis labels and descriptive captions. Tables, if any, must have descriptive captions.
5. This document may be updated if a mistake is found or further clarification is needed. A Blackboard announcement will be posted as soon as this document is updated.
6. Submission guidelines.
 - (a) Label your report as ENME408_S26_Project02_<lastnames>.pdf.
 - (b) Include all figures and tables with captions in your report.
 - (c) You may include MATLAB scripts and representative outputs as Appendices. Label the appendices and cite them in your report appropriately.
 - (d) Place all TEX files, MATLAB scripts, and Simulink models in a folder, then archive it. Label your archive as ENME408_S26_Project02_<lastnames>.zip.
 - (e) Upload your pdf file and the archive folder to Blackboard before the deadline.
7. Honor code: "I have neither received nor given aid to anyone outside my team during this project."

Sign/print your names here.

If you cannot sign/print in the box above, please write the honor code on the first page of your submission and sign it.

8. Reports with substantial similarity will be reported to the UMBC Academic Conduct Committee.
9. Include a **credit author statement** in your report. The statement recognizes individual author contributions and facilitates collaboration. For example,
Author 1: Methodology, Software, Visualization
Author 2: Data curation, Writing, Original draft preparation, Validation.

Attitude Stabilization, Position Control and Tracking

In this mini-project, you will design and evaluate feedback controllers for the quadrotor attitude and translational dynamics, and integrate them with a simple guidance layer for waypoint and trajectory tracking. The objective is to move from open-loop dynamics (Proj 01) to closed-loop stabilization and guidance under realistic constraints.

1. **Linearization and attitude dynamics.** The *full nonlinear* multicopter equations of motion are

$$m\ddot{p} = mge_3 + \mathcal{O}_{A/D}e_3f, \quad (1)$$

$$\dot{q} = S(q)^{-1}\omega, \quad (2)$$

$$J\dot{\omega} = \omega^\times J\omega + \tau. \quad (3)$$

Note that (1) is the translational dynamics and (2)-(3) is the rotational dynamics. Define $x_{\text{att}} = [\phi, \theta, \psi, \omega_1, \omega_2, \omega_3]^T$ and $u_{\text{att}} = [\tau_1, \tau_2, \tau_3]^T$.

- (a) Linearize the rotational equations of motion about the hover equilibrium $x_{\text{att}} = 0_{6 \times 1}$ and $u_{\text{att}} = 0_{3 \times 1}$ to obtain the linearized attitude dynamics in state-space form

$$\dot{x}_{\text{att}} = A_{\text{att}}x_{\text{att}} + B_{\text{att}}u_{\text{att}}. \quad (4)$$

- (b) Verify controllability of the linearized attitude subsystem.

2. **Full-state feedback attitude stabilization.**

- (a) Design a full-state feedback controller

$$u_{\text{att}} = K_{\text{att}}x_{\text{att}} \quad (5)$$

using pole placement or LQR.

- (b) Simulate the closed-loop nonlinear attitude dynamics using the full nonlinear rotational EOM.
 (c) Use a set of nonzero initial conditions (e.g., $\phi(0), \theta(0), \psi(0), \omega(0)$) to demonstrate stabilization to hover.
 (d) Plot Euler angles and angular velocities versus time, and comment on transient response (rise time, overshoot, settling time).

3. **Region of attraction via initial-condition sweep in (ϕ, θ) .**

- (a) **Grid/box of initial conditions.** Consider only roll and pitch initial conditions, with yaw fixed at zero, that is,

$$\psi(0) = 0, \quad \omega(0) = 0, \quad \phi(0) \in [-85^\circ, 85^\circ], \quad \theta(0) \in [-85^\circ, 85^\circ]. \quad (6)$$

Generate a uniform grid of $(\phi(0), \theta(0))$ pairs over this box (choose a reasonable resolution, e.g., 5° increments). For each grid point, simulate the *nonlinear* closed-loop attitude dynamics using your full-state feedback controller.

- (b) **Convergence criterion.** Define an initial condition as *converged* if, for a chosen simulation horizon T ,

$$\max_{t \in [T-\Delta, T]} (|\phi(t)|, |\theta(t)|) \leq \varepsilon, \quad (7)$$

where ε is a small angle tolerance (e.g., $\varepsilon = 2^\circ$) and $\Delta = 0.5$ s is a short terminal window. Otherwise, define it as *non-convergent*. (You may also label an initial condition as *non-convergent* if numerical issues occur or if angles grow without bound.)

- (c) **Region of attraction plot.** Create a scatter plot in the $(\phi(0), \theta(0))$ plane, where the

- convergent initial conditions are plotted in **green**.
- non-convergent initial conditions are plotted in **red**.

Label axes in degrees and include a descriptive caption. This plot is an empirical approximation of the region of attraction for the nonlinear closed-loop system (restricted to $\psi(0) = 0$ and $\omega(0) = 0$).

- (d) **Discussion.** Briefly discuss the observed shape/extent of the convergent region and explain why a controller fails for large initial angles.

4. **Translational dynamics and position control with integral action.** The translational dynamics of the multicopter are given by (1), where f is the total thrust magnitude. Note that the thrust direction $\mathcal{O}_{A/D}e_3$ depends on the attitude of the multicopter and is therefore state-dependent and nonlinear.

To simplify the controller design for the translational subsystem, we adopt a commonly used cascaded-control approximation, assuming the attitude dynamics are sufficiently fast relative to the translational dynamics. Under this assumption, the thrust direction can be treated as an independently commandable input through appropriate attitude control. In particular, consider the idealized translational model

$$m\ddot{p} = mge_3 + \bar{f}, \quad (8)$$

where $\bar{f} \in \mathbb{R}^3$ is a *virtual force* command generated by the position controller. Note that the idealized model is linear.

- Design a full-state feedback controller with integral action for the idealized translational model to achieve position regulation and tracking.
- Let $p_{\text{des}} = e_1$. Simulate the idealized translational dynamics with the controller in closed loop and plot all states.
- Plot the 3D trajectory using `plot3` and overlay virtual force vector $\bar{f}$ using `quiver3`.

Remark 1. The resulting virtual force command $\bar{f}(t)$ must then be mapped to physically realizable control commands by computing appropriate roll and pitch reference angles and thrust magnitude. These references are to be tracked by the attitude controller designed in the previous step.

Remark 2. The above design implicitly relies on a time-scale separation between the inner-loop attitude dynamics and the outer-loop translational dynamics. In practice, this assumption may be violated. In particular, aggressive position commands or poorly tuned attitude controllers may violate this condition, leading to degraded tracking performance or instability.

5. **Combined translational–rotational control (cascaded implementation).** In this step, you will combine your outer-loop translational controller (virtual force $\bar{f}$) with your inner-loop attitude controller (torque input τ) to control the *full nonlinear* multicopter dynamics.

- (a) **Virtual-force to attitude/thrust mapping.** At each time t , your translational controller produces a virtual force command $\bar{f}(t) \in \mathbb{R}^3$ from the idealized model

$$m\ddot{p} = mge_3 + \bar{f}. \quad (9)$$

Convert $\bar{f}(t)$ into a thrust magnitude command $f(t)$ and roll/pitch reference angles $(\phi_{\text{ref}}(t), \theta_{\text{ref}}(t))$ by enforcing the matching condition

$$\mathcal{O}_{A/D}(\phi_{\text{ref}}, \theta_{\text{ref}}, \psi_{\text{ref}}) e_3 f = \bar{f}. \quad (10)$$

For this project, set $\psi_{\text{ref}}(t) = 0$ for all t .

- (b) **Inner-loop attitude tracking.** Modify your attitude controller so that it stabilizes the attitude *about the time-varying reference* $(\phi_{\text{ref}}(t), \theta_{\text{ref}}(t), \psi_{\text{ref}}(t))$ instead of the origin. In particular, define attitude and angular-rate error states (e.g., $e_\phi = \phi - \phi_{\text{ref}}$, $e_\theta = \theta - \theta_{\text{ref}}$, $e_\psi = \psi - \psi_{\text{ref}}$, and rate errors) and use your full-state feedback law to generate the torque command

$$\tau(t) = K_{\text{att}} x_{\text{att,err}}(t). \quad (11)$$

- (c) **Full nonlinear simulation.** Simulate the *full nonlinear* multicopter equations of motion (1)-(3). Use your cascaded controller to regulate the position to

$$p_{\text{des}} = e_1. \quad (12)$$

- (d) **Plots and performance metrics.** Include the following plots

- $p(t)$ and $\dot{p}(t)$ versus time, and the position tracking error $p(t) - p_{\text{des}}$.
- $\phi(t)$ and $\theta(t)$ versus time, along with $\phi_{\text{ref}}(t)$ and $\theta_{\text{ref}}(t)$ on the same plots.
- Thrust command $f(t)$ and torque commands $\tau_1(t), \tau_2(t), \tau_3(t)$ versus time.
- A 3D trajectory plot using `plot3`. Optionally overlay the commanded thrust direction $\mathcal{O}_{A/D}(\phi_{\text{ref}}, \theta_{\text{ref}}, 0)e_3$ using `quiver3`.

Comment on transient response, steady-state error, and how well the inner-loop tracks the commanded references.

- (e) **Stress test (cascaded assumption).** Increase the aggressiveness of the position controller (for example, by assigning faster closed-loop poles / larger gains) and repeat the simulation. Identify when the cascaded assumption begins to degrade performance (e.g., overshoot, oscillations, loss of tracking).

Remark 3. This step implements a standard cascaded architecture: the outer loop computes a desired force direction (via $\hat{f}$), while the inner loop attempts to realize that direction by tracking $(\phi_{\text{ref}}, \theta_{\text{ref}})$ using torque commands. The overall closed-loop performance depends critically on the relative bandwidths of the two loops.

Obtaining Desired Pitch and Roll from the Desired Force Vector

Note that

$$\mathcal{O}_{A/D}e_3 = \frac{f}{\|f\|} = \hat{f} \quad (13)$$

which implies that

$$\mathcal{O}_3^T(\psi)\mathcal{O}_2^T(\theta)\mathcal{O}_1^T(\phi)e_3 = \hat{f}. \quad (14)$$

Since $\psi(t)$ is user specified, it follows that

$$\mathcal{O}_2^T(\theta)\mathcal{O}_1^T(\phi)e_3 = \mathcal{O}_3(\psi)\hat{f} = \bar{f}, \quad (15)$$

which can be expanded to

$$\begin{bmatrix} \cos \theta & 0 & \sin \theta \\ 0 & 1 & 0 \\ -\sin \theta & 0 & \cos \theta \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \phi & -\sin \phi \\ 0 & \sin \phi & \cos \phi \end{bmatrix} e_3 = \begin{bmatrix} \bar{f}_1 \\ \bar{f}_2 \\ \bar{f}_3 \end{bmatrix} = \begin{bmatrix} \sin \theta \cos \phi \\ -\sin \phi \\ \cos \theta \cos \phi \end{bmatrix}, \quad (16)$$

which implies that

$$\theta = \text{atan2}(\bar{f}_1, \bar{f}_3), \quad (17)$$

$$\phi = \text{atan2}(-\bar{f}_2, \sqrt{\bar{f}_1^2 + \bar{f}_3^2}). \quad (18)$$

Remark: Although ϕ can be computed as $\phi = -\sin^{-1}(\bar{f}_2)$, the expression using $\text{atan2}(\cdot, \cdot)$ is preferred in practice. This is because $\sin^{-1}(\cdot)$ returns values only in $[-\pi/2, \pi/2]$ and does not distinguish between configurations with the same sine but different cosine. In contrast, $\text{atan2}(\cdot, \cdot)$ incorporates both sine and cosine information, ensures the correct quadrant of ϕ , and provides improved numerical robustness, particularly when $\cos \phi \approx 0$.